

For M_2 to be in saturation

$$V_{DS} \geq V_{GS} - V_{th2}$$

$$V_X - V_b \geq V_{out} - V_b - V_{th2}$$

$$V_X + V_{th2} \geq V_{out}$$

$$V_{b1} - V_{th4} + V_{th2} \geq V_{out} \quad \text{--- (1)}$$

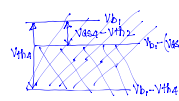
For M_4 to be in saturation

$$V_{DS} \geq V_{GS} - V_{th4}$$

$$V_{out} - V_X \geq V_{b1} - V_X - V_{th4}$$

$$V_{out} \geq V_{b1} - V_{th4} \quad \text{--- (2)}$$

$$V_{b1} - V_{th4} \leq V_{out} \leq V_{b1} - V_{th4} + V_{th2}$$



Design of telescopic amplifier :-

- ① I_D ② V_{ov} ③ $\frac{W}{L}$ ④ g_m ⑤ I_o

$$V_{DD} = 3V$$

$$\text{Power Budget} = 10 \text{ mW}$$

$$\text{Open loop gain} = 2000 \text{ V/V}$$

$$\text{Output voltage swing} = 2V \text{ peak to peak}$$

$$\mu_{n\text{ox}} = 60 \text{ cm}^2/\text{Vs}$$

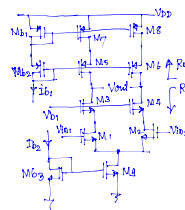
$$\mu_{p\text{ox}} = 20 \text{ cm}^2/\text{Vs}$$

$$\gamma = 0$$

$$\lambda_n = 0.1 \text{ V}^{-1} \quad \lambda_p = 0.2 \text{ V}^{-1} \quad L = 0.5 \mu\text{m}$$

$$\gamma = 0$$

$$V_{thn} = |V_{thp}| = 0.7V$$



$$P_D = V_{DD} \times I_{\text{total}}$$

$$10 \text{ mW} = 3V \times I_{\text{total}}$$

$$I_{\text{total}} = 3.33 \text{ mA}$$

$$I_{\text{total}} = I_{b1} + I_{b2} + I_{b3} + I_{b4}$$

$$= 0.33 \text{ mA} + 2 \text{ mA}$$

$$I_{D7} = I_{D6} = 1.5 \text{ mA}$$

$$\text{Output voltage swing} = 2 \left[V_{DD} - \left(|V_{ovs}| + |V_{ovs}| + V_{ov4} + V_{ov2} + V_{ov3} \right) \right]$$

$$3 = 2 \left[3 - \left(|V_{ovs}| + |V_{ovs}| + V_{ov4} + V_{ov2} + V_{ov3} \right) \right]$$

$$|V_{ovs}| + |V_{ovs}| + V_{ov4} + V_{ov2} + V_{ov3} = 1.5V$$

$$\text{Let } V_{ov3} = 0.5V$$

$$|V_{ovs}| = |V_{ovs}| = 0.2V$$

$$V_{ov4} = V_{ov2} = 0.2V$$

$$I_{D1} = \frac{1}{2} \mu_{n\text{ox}} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$1.5 \text{ mA} = \frac{1}{2} \times 60 \times 10^{-6} \times \frac{W}{L} (0.2)^2$$

$$\left(\frac{W}{L} \right)_{1-4} = 1250 \text{ V}^2$$

$$I_{D7} = \frac{1}{2} \mu_{p\text{ox}} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$1.5 \text{ mA} = \frac{1}{2} \times 20 \times 10^{-6} \times \frac{W}{L} (0.3)^2$$

$$\left(\frac{W}{L} \right)_{5-8} = 1111 \text{ V}^2 = \left(\frac{555}{0.5} \right)^2 \approx \left(\frac{1111}{1.2} \right)^2$$

$$\left(\frac{W}{L} \right)_9 = 400$$

$$A_v = g_{m12} [R_{up} || R_{down}] = 15 \text{ ms} \left[\frac{111 \text{ K} || 666 \text{ K}}{1} \right] = 1428 \text{ V/V}$$

$$R_{up} = g_{m5} V_{ov5} \rightarrow 111 \text{ K}\Omega$$

$$R_{down} = g_{m12} V_{ov2} = 666 \text{ K}\Omega$$

$$g_{m6} = \frac{2I_{D6}}{V_{ov6}} = \frac{2 \times 1.5 \text{ mA}}{0.2} = 15 \text{ mS}$$

$$V_{ov5} = V_{ov6} = \frac{1}{\lambda_p I_{D5}} = \frac{1}{0.2 \times 1.5 \text{ mA}} = 3.33 \text{ V}$$

$$g_{m4} = \frac{2I_{D4}}{V_{ov4}} = \frac{2 \times 1.5 \text{ mA}}{0.2} = 15 \text{ mS}$$

$$V_{ov4} = \frac{1}{\lambda_n I_{D4}} = \frac{1}{0.1 \times 1.5 \text{ mA}} = 6.66 \text{ V}$$

$$R_{up} = g_{m5} V_{ov5}$$

$$g_{m12} = \sqrt{2 \mu_{p\text{ox}} \frac{W}{L} I_{D12}} \times \frac{1}{\lambda_p V_{ov}}$$

$$\propto \sqrt{\frac{W}{L}} \times \frac{1}{I_{D12}}$$

$$\propto \sqrt{\frac{W}{L}} \times \frac{1}{I_{D12}}$$

$$\lambda \propto \frac{1}{L}$$

$$\text{For } L = 1 \mu\text{m} \text{ for } M5 - M8$$

$$\lambda \propto \frac{1}{L}$$

$$\text{If } L \text{ is doubled, } \lambda \text{ becomes half}$$

$$\therefore \lambda_p = 0.1 \text{ V}^{-1}$$

$$V_{ov5} = V_{ov6} = \frac{1}{\lambda_p I_{D5}} = \frac{1}{0.1 \times 1.5 \text{ mA}} = 6.67 \text{ V}$$

$$R_{up} = g_{m5} V_{ov5} = 10 \text{ ms} \times 6.67 \text{ V} \times 6.67 \text{ K}\Omega = 444.89 \text{ K}\Omega$$

$$A_v = 15 \text{ ms} [444.89 \text{ K}\Omega || 666 \text{ K}\Omega]$$

$$\approx 4000 \text{ V/V}$$

which meets the gain specifications